Architecture of Setup and config Wazuh

- Setup Virtual Environment
  - Wazuh Server
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Setup Virtual Environment

Wazuh Server

- IP check

```
┌──(root㉿kali)-[~]
└─# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:a3:3f:42 brd ff:ff:ff:ff:ff:ff
    inet 192.168.35.128/24 brd 192.168.35.255 scope global dynamic noprefixroute eth0
       valid_lft 1173sec preferred_lft 1173sec
    inet6 fe80::f1bb:f476:547d:de8c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(root㉿kali)-[~]
└─# ip route
default via 192.168.35.1 dev eth0 proto dhcp src 192.168.35.128 metric 100
192.168.35.0/24 dev eth0 proto kernel scope link src 192.168.35.128 metric 100
```

Wazuh Agent

- IP check

```
┌──(root㉿kali)-[~]
└─# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:a1:ff:61 brd ff:ff:ff:ff:ff:ff
    inet 192.168.35.129/24 brd 192.168.35.255 scope global dynamic noprefixroute eth0
       valid_lft 1758sec preferred_lft 1758sec
    inet6 fe80::c534:e757:f402:b6d3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(root㉿kali)-[~]
└─# ip route
default via 192.168.35.1 dev eth0 proto dhcp src 192.168.35.129 metric 100
192.168.35.0/24 dev eth0 proto kernel scope link src 192.168.35.129 metric 100
```

Wazuh Attacker

- IP check

```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
1000
    link/ether 00:0c:29:e6:b2:6c brd ff:ff:ff:ff:ff:ff
    inet 192.168.35.141/24 brd 192.168.35.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fee6:b26c/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever

┌──(kali㉿kali)-[~]
└─$ ip route
default via 192.168.35.1 dev eth0 onlink
192.168.35.0/24 dev eth0 proto kernel scope link src 192.168.35.141
```

Check Connection

From Wazuh Agent to Wazuh Server and Attacker

```
┌──(root㉿kali)-[~]
└─# ping 192.168.35.128
PING 192.168.35.128 (192.168.35.128) 56(84) bytes of data.
64 bytes from 192.168.35.128: icmp_seq=1 ttl=64 time=16.6 ms
64 bytes from 192.168.35.128: icmp_seq=2 ttl=64 time=0.726 ms
64 bytes from 192.168.35.128: icmp_seq=3 ttl=64 time=0.367 ms
64 bytes from 192.168.35.128: icmp_seq=4 ttl=64 time=1.81 ms
64 bytes from 192.168.35.128: icmp_seq=5 ttl=64 time=2.83 ms
^Z
zsh: suspended  ping 192.168.35.128

┌──(root㉿kali)-[~]
└─# ping 192.168.35.141
PING 192.168.35.141 (192.168.35.141) 56(84) bytes of data.
64 bytes from 192.168.35.141: icmp_seq=1 ttl=64 time=1.13 ms
64 bytes from 192.168.35.141: icmp_seq=2 ttl=64 time=0.423 ms
64 bytes from 192.168.35.141: icmp_seq=3 ttl=64 time=0.303 ms
64 bytes from 192.168.35.141: icmp_seq=4 ttl=64 time=0.341 ms
^Z
zsh: suspended  ping 192.168.35.141
```

From Attacker to Wazuh Server and Agent

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.35.128
PING 192.168.35.128 (192.168.35.128) 56(84) bytes of data.
64 bytes from 192.168.35.128: icmp_seq=1 ttl=64 time=16.1 ms
64 bytes from 192.168.35.128: icmp_seq=2 ttl=64 time=0.340 ms
64 bytes from 192.168.35.128: icmp_seq=3 ttl=64 time=0.342 ms
64 bytes from 192.168.35.128: icmp_seq=4 ttl=64 time=10.6 ms
^Z
zsh: suspended  ping 192.168.35.128

┌──(kali㉿kali)-[~]
└─$ ping 192.168.35.129
PING 192.168.35.129 (192.168.35.129) 56(84) bytes of data.
64 bytes from 192.168.35.129: icmp_seq=1 ttl=64 time=0.657 ms
64 bytes from 192.168.35.129: icmp_seq=2 ttl=64 time=0.563 ms
64 bytes from 192.168.35.129: icmp_seq=3 ttl=64 time=0.379 ms
64 bytes from 192.168.35.129: icmp_seq=4 ttl=64 time=0.465 ms
^Z
zsh: suspended  ping 192.168.35.129
```

Installing and Configuring Wazuh Components

Wazuh server

```
sudo apt update && sudo apt upgrade -y
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
sudo bash wazuh-install.sh -a
```

- Status Wazuh-manager

```
● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
     Active: active (running) since Wed 2025-11-26 23:19:11 EST; 20min ago
 Invocation: 0a45874f934e48f990bdf196fb801f2e
      Tasks: 231 (limit: 4410)
     Memory: 1.2G (peak: 1.4G, swap: 149.1M, swap peak: 204.3M)
        CPU: 3min 51.495s
     CGroup: /system.slice/wazuh-manager.service
             ├─140012 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─140013 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─140014 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─140017 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─140020 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             ├─140061 /var/ossec/bin/wazuh-authd
             ├─140084 /var/ossec/bin/wazuh-db
             ├─140127 /var/ossec/bin/wazuh-execd
             ├─140153 /var/ossec/bin/wazuh-analysisd
             ├─140230 /var/ossec/bin/wazuh-syscheckd
             ├─140254 /var/ossec/bin/wazuh-remoted
             ├─140298 /var/ossec/bin/wazuh-logcollector
             ├─140325 /var/ossec/bin/wazuh-monitord
             └─140356 /var/ossec/bin/wazuh-modulesd

Nov 26 23:19:05 kali env[139886]: Started wazuh-syscheckd...
Nov 26 23:19:06 kali env[139886]: Started wazuh-remoted...
Nov 26 23:19:07 kali env[139886]: Started wazuh-logcollector...
Nov 26 23:19:08 kali env[139886]: Started wazuh-monitord...
Nov 26 23:19:08 kali env[140353]: 2025/11/26 23:19:08 wazuh-modulesd:router: INFO: Loaded router module.
Nov 26 23:19:08 kali env[140353]: 2025/11/26 23:19:08 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Nov 26 23:19:08 kali env[140353]: 2025/11/26 23:19:08 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester mo>
Nov 26 23:19:09 kali env[139886]: Started wazuh-modulesd...
Nov 26 23:19:11 kali env[139886]: Completed.
Nov 26 23:19:11 kali systemd[1]: Started wazuh-manager.service - Wazuh manager.
```

- Status Wazuh-indexer

```
● wazuh-indexer.service - wazuh-indexer
     Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: disabled)
     Active: active (running) since Wed 2025-11-26 23:07:06 EST; 32min ago
 Invocation: 6f6e34779bb741c0acc497365065eb3a
       Docs: https://documentation.wazuh.com
   Main PID: 79927 (java)
      Tasks: 88 (limit: 4410)
     Memory: 1.1G (peak: 1.3G, swap: 118.2M, swap peak: 133.1M)
        CPU: 1min 18.693s
     CGroup: /system.slice/wazuh-indexer.service
             └─79927 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch>

Nov 26 23:06:57 kali systemd-entrypoint[79927]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstr>
Nov 26 23:06:57 kali systemd-entrypoint[79927]: WARNING: Please consider reporting this to the maintainers of org.opensearch.>
Nov 26 23:06:57 kali systemd-entrypoint[79927]: WARNING: System::setSecurityManager will be removed in a future release
Nov 26 23:06:58 kali systemd-entrypoint[79927]: Nov 26, 2025 11:06:58 PM sun.util.locale.provider.LocaleProviderAdapter <clin>
Nov 26 23:06:58 kali systemd-entrypoint[79927]: WARNING: COMPAT locale provider will be removed in a future release
Nov 26 23:06:58 kali systemd-entrypoint[79927]: WARNING: A terminally deprecated method in java.lang.System has been called
Nov 26 23:06:58 kali systemd-entrypoint[79927]: WARNING: System::setSecurityManager has been called by org.opensearch.bootstr>
Nov 26 23:06:58 kali systemd-entrypoint[79927]: WARNING: Please consider reporting this to the maintainers of org.opensearch.>
Nov 26 23:06:58 kali systemd-entrypoint[79927]: WARNING: System::setSecurityManager will be removed in a future release
Nov 26 23:07:06 kali systemd[1]: Started wazuh-indexer.service - wazuh-indexer.
```

- Status Wazuh-dashboard

```
● wazuh-dashboard.service - wazuh-dashboard
     Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; preset: disabled)
     Active: active (running) since Wed 2025-11-26 23:19:24 EST; 20min ago
 Invocation: 7c5aaeadd0d04b249f662b944b0d30ff
   Main PID: 142278 (node)
      Tasks: 11 (limit: 4410)
     Memory: 226M (peak: 269.7M)
        CPU: 16.225s
     CGroup: /system.slice/wazuh-dashboard.service
             └─142278 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65536 --unhandled-rejecti>

Nov 26 23:21:18 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:21:18Z","tags":["info","plugins>
Nov 26 23:21:18 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:21:18Z","tags":["info","plugins>
Nov 26 23:21:18 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:21:18Z","tags":["listening","in>
Nov 26 23:21:18 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:21:18Z","tags":["info","http","">
Nov 26 23:21:19 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:21:19Z","tags":["info","plugins>
Nov 26 23:21:25 kali opensearch-dashboards[142278]: [agentkeepalive:deprecated] options.freeSocketKeepAliveTimeout is depreca>
Nov 26 23:21:32 kali opensearch-dashboards[142278]: {"type":"response","@timestamp":"2025-11-27T04:21:25Z","tags":[],"pid":14>
Nov 26 23:25:00 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:25:00Z","tags":["error","opense>
Nov 26 23:25:00 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:25:00Z","tags":["info","plugins>
Nov 26 23:30:00 kali opensearch-dashboards[142278]: {"type":"log","@timestamp":"2025-11-27T04:30:00Z","tags":["info","plugins>
```

- Status filebeat

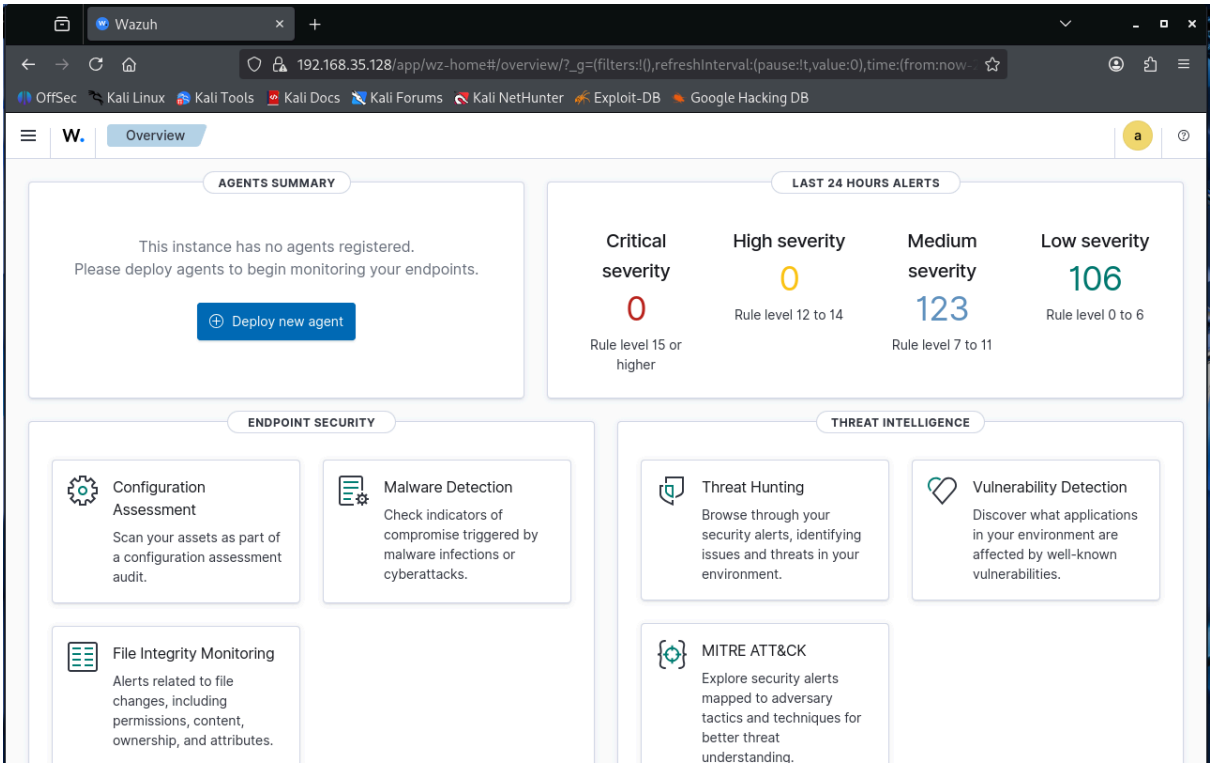
```
● filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
     Loaded: loaded (/usr/lib/systemd/system/filebeat.service; enabled; preset: disabled)
     Active: active (running) since Wed 2025-11-26 23:18:42 EST; 20min ago
 Invocation: 846dd85b103a44b2b9757797533cbdf2
       Docs: https://www.elastic.co/products/beats/filebeat
   Main PID: 139585 (filebeat)
      Tasks: 9 (limit: 4410)
     Memory: 15.8M (peak: 16M)
        CPU: 309ms
     CGroup: /system.slice/filebeat.service
             └─139585 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebeat/filebeat.yml --path.home /usr/s>

Nov 26 23:18:42 kali systemd[1]: Started filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch>
lines 54-92/92 (END)
```

- After setup of Wazuh Server, it provided the User name and Password and how to access to the website

```
26/11/2025 23:21:51 INFO: --- Summary ---
26/11/2025 23:21:51 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password: v5laOMLF25J.Uw0QOQIE9rdJI9zv*rV7
26/11/2025 23:21:51 INFO: Installation finished.
```

- As we can see this is the web interface of wazuh



Wazuh Agent

```
sudo apt update
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --dearmo
r -o /usr/share/keyrings/wazuh.gpg
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.w
azuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wazuh.list
sudo apt update
```

- Use curl to silently download Wazuh GPG key and pipe it into —dearmor to convert the key from ASCII to binary format and save the binary key to /usr/share/keyrings/wazuh.gpg. This command is used to verify the authenticity of packages from Wazuh's repository
- After the stage of download and store Wazuh's GPG key, then added the wazuh's official repository to the system by create a new repository entry for wazuh and specifies that packages from this repository must be verified using the GPG key stored at /usr/share/keyrings/wazuh.gpg

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --dearmor -o /usr/share/keyrings/wazuh.gpg
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wazuh.list
sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main
Get:1 https://packages.wazuh.com/4.x/apt stable InRelease [17.3 kB]
Get:2 https://packages.wazuh.com/4.x/apt stable/main amd64 Packages [49.4 kB]
Get:3 https://packages.wazuh.com/4.x/apt stable/main amd64 Contents (deb) [2,022 kB]
Hit:4 http://http.kali.org/kali kali-rolling InRelease
Fetched 2,088 kB in 1s (2,238 kB/s)
6 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(root㉿kali)-[~]
└─#
```

```
sudo WAZUH_MANAGER="192.168.35.128" apt install wazuh-agent -y

sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

- The environment variable WAZUH_MANAGER="192.168.35.138" tells the installer which server (Manager) the agent should connect to and during installation, configuration file (/var/ossec/etc/ossec.conf) is updated with this IP address. Without this, the agent wouldn't know where to send logs and alerts. By setting the Manager IP here, you avoid editing the config file manually later.
- Next step is that register the Wazuh Agent service and will automatically start
- Finally, run systemctl start to launches the agent with several processes start
  - `wazuh-execd` → executes commands from the Manager.
  - `wazuh-agentd` → handles communication with the Manager.
  - `wazuh-syscheckd` → performs file integrity monitoring.
  - `wazuh-logcollector` → collects logs from the system.
  - `wazuh-modulesd` → runs extra modules like vulnerability detection.

```
┌──(root㉿kali)-[~]
└─# sudo WAZUH_MANAGER="192.168.35.139" apt install wazuh-agent -y

sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Installing:
  wazuh-agent

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 6
  Download size: 13.1 MB
  Space needed: 48.4 MB / 62.5 GB available

Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-agent amd64 4.14.1-1 [13.1 MB]
Fetched 13.1 MB in 2s (5,556 kB/s)
Preconfiguring packages ...
Selecting previously unselected package wazuh-agent.
(Reading database ... 432741 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.1-1_amd64.deb ...
Unpacking wazuh-agent (4.14.1-1) ...
Setting up wazuh-agent (4.14.1-1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.
```

- As we can see, i have configured wrong ip so I will change it in /var/ossec/etc/ossec.conf with the ip is 192.168.35.128 and restart the agent again
- At this point, we need to enroll the agent with the Manager with command on Wazuh-Server

```
sudo /var/ossec/bin/manage_agents
```

- Enter the new agent, assign with name: wazuh-agent, ip: 192.168.35.129

```
└─# sudo /var/ossec/bin/manage_agents



****************************************
* Wazuh v4.14.1 Agent manager.         *
* The following options are available: *
****************************************
   (A)dd an agent (A).
   (E)xtract key for an agent (E).
   (L)ist already added agents (L).
   (R)emove an agent (R).
   (Q)uit.
Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
  Please provide the following:
   * A name for the new agent: wazuh-agent
   * The IP Address of the new agent: 192.168.35.129
Confirm adding it?(y/n): y
Agent added with ID 001.
```

- The Manager generates a unique key for this agent, we can see in the picture

```
****************************************
* Wazuh v4.14.1 Agent manager.         *
* The following options are available: *
****************************************
   (A)dd an agent (A).
   (E)xtract key for an agent (E).
   (L)ist already added agents (L).
   (R)emove an agent (R).
   (Q)uit.
Choose your action: A,E,L,R or Q: E

Available agents:
   ID: 001, Name: wazuh-agent, IP: 192.168.35.129
Provide the ID of the agent to extract the key (or '\q' to quit): 001

Agent key information for '001' is:
MDAxIHdhenVoLWFnZW50IDE5Mi4xNjguMzUuMTI5IDY3NThhMjRiOGE0YTA2ZDgwNzJjOGU1NDg3N2NmZmFkYzBlODdjNDhiODNlMjFhNzkxNmVkZjJlMzVhMWY4MD
I=
```

- Next, on the side of Wazuh-Agent, i use this command to import the key and then restart

```
sudo /var/ossec/bin/manage_agents
```

- As the results, we know can see the lists all agents known to the Manager by this command on server

```
sudo /var/ossec/bin/agent_control -l
```

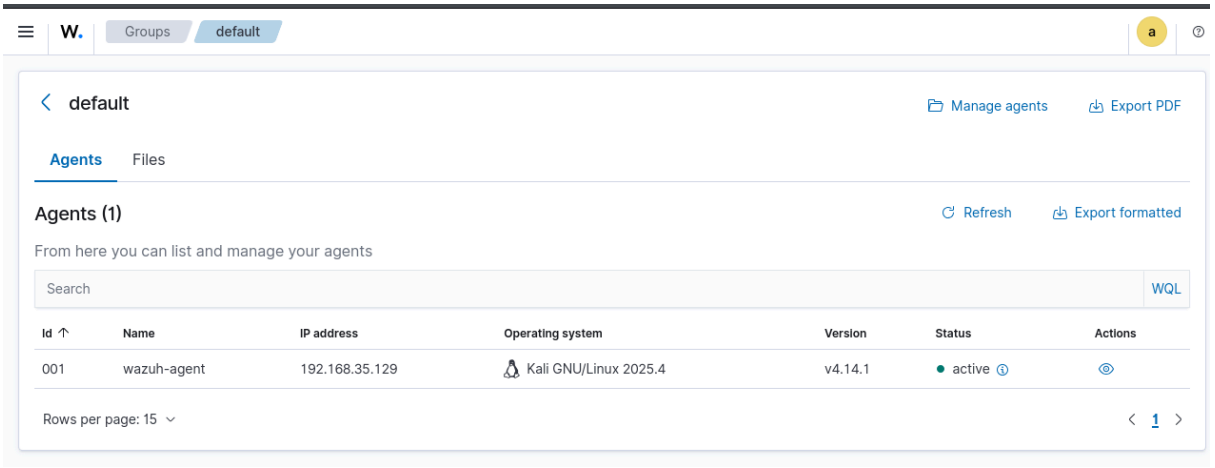
```
┌──(root㉿kali)-[~]
└─# sudo /var/ossec/bin/agent_control -l


Wazuh agent_control. List of available agents:
   ID: 000, Name: kali (server), IP: 127.0.0.1, Active/Local
   ID: 001, Name: wazuh-agent, IP: 192.168.35.129, Active

List of agentless devices:


┌──(root㉿kali)-[~]
└─#
```

- Or we call see the agent on dashboard



Usecase

[USECASE 5.1: BLOCKING ROTATING IP ATTACK]

[USECASE 5.1: SSH Multiple Source IP Brute-Force Attack - BASIC]

[USECASE 5.2: INSIDER THREAT - UNAUTHORIZED SUDO PRIVILEGE ESCALATION]

[USECASE 5.3: CRON JOB PERSISTENCE DETECTION]

Advance

[USECASE 5.3: WEBSHELL UPLOAD & DETECTION]

[USECASE 5.4: CRYPTOCURRENCY MINER DETECTION]

[Untitled]